
Part 1 — Programming the data plane with DOCA Flow

Programming SmartNICs: From Packet Processing to Programmable Transport

ACM SIGCOMM 2026

Fernando Ramos  Muhammad Shahbaz  Mina Tahmasbi Arashloo  Mario Baldi 

Francisco Pereira  Bilal Saleem  Tyler Liu  Chenxingyu Zhao 

 INESC-ID, Instituto Superior Técnico, University of Lisbon  University of Michigan  University of Waterloo
 NVIDIA  Purdue University  University of Washington

In this part you program the **data plane** of a BlueField-3: you tell the NIC what to do with packets *in its own hardware, at line rate*, before any CPU sees them. You will complete a program that marks live RoCE traffic with an ECN congestion signal — the signal the congestion controller you write in Part 2 reacts to.

Prerequisites. You are logged into the **Arm cores** of a BlueField-3 — a normal Ubuntu shell that happens to run inside the NIC. You have this repository in your home directory and can run `sudo`. **Every command below is typed on the Arm cores**; the host is never involved.

First, check which DOCA release your card runs, because the exercise ships in two versions:

```
$ cat /opt/mellanox/doca/applications/VERSION 2>/dev/null || cat /opt/mellanox/doca/VERSION
```

DOCA 2.7 or 2.9 means you work in `doca-2/`; DOCA 3.1 or 3.4 means `doca-3/`. This guide marks the few places they differ with `[2.x]` and `[3.x]`; everything unmarked applies to both.

Part A — The card, and getting traffic onto it

Your card has **two ports connected to each other**, so whatever leaves `p1` arrives at `p0`. That makes a single card behave like a small two-node network talking to itself. The two endpoints are lightweight virtual NICs called **SFs** (sub-functions), one per port, each in its own network namespace so the kernel cannot short-circuit them locally:

Endpoint	RDMA device	Namespace	IP	Role
SF on port 0	<code>m1x5_2</code>	<code>ns0</code>	<code>10.0.0.1</code>	server (receives)
SF on port 1	<code>m1x5_3</code>	<code>ns1</code>	<code>10.0.0.2</code>	client (sends)

The client sends RDMA WRITES to the server. They leave `p1`, cross the cable, and arrive at `p0` — where **your program** sees them.

What you are reproducing is an everyday situation in a datacenter network: a sender is going as fast as it can, the network becomes congested, a switch on the path signals that congestion by setting the ECN bits in the IP header, and the sender slows down. Here, **your card plays the congested switch** — no congestion actually exists, you are *manufacturing* the signal, which is what makes it easy to see and to dial up and down.

That splits into the two halves of the tutorial:

- **Part 1, this guide.** Program the NIC to set the ECN “congestion experienced” (CE) mark on a fraction of the packets going past — you choose the fraction — and to copy a sample of them to a file so you can check your work. Everything happens in hardware; your program only installs the rules.
- **Part 2.** Program the DPA to *react*: the server’s NIC answers a CE-marked packet with a congestion notification, and your algorithm turns each one into a lower send rate for the client.

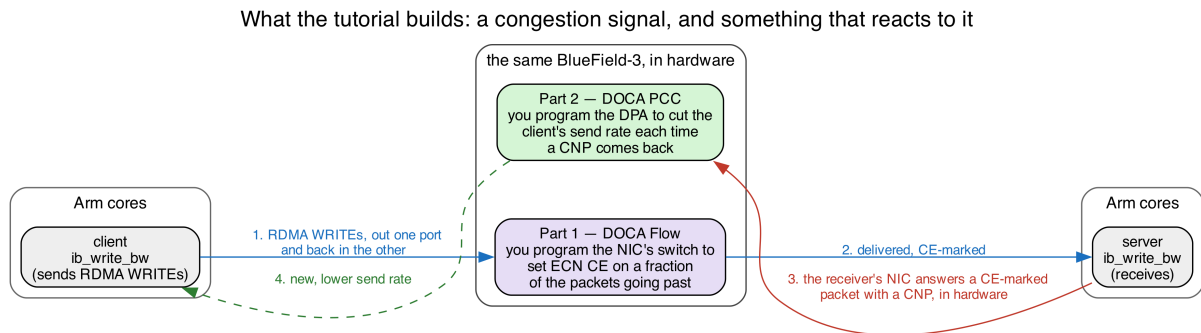


Figure 1: You write the marking in Part 1 and the reaction in Part 2. The client, the server, and the CNP the receiver’s NIC sends back are already there.

Where things are. The repository has one directory per DOCA release, plus a matching `-solutions` directory holding the finished program:

```
doca-2/doca-flow/doca_flow_ecn_pcap.c    <- the file you edit
doca-2-solutions/doca-flow/doca_flow_ecn_pcap.c  <- the finished version
scripts/                                     <- traffic generators
```

Substitute `doca-3` throughout if that is your release. Every command in this guide is run from the top of the repository, so nothing below needs a `cd`.

The network devices you will use. Run this to see the card’s network interfaces:

```
$ ip -br link show | grep -E '^p0|^p1|^enp3'
p0          UP    ...    # physical port 0
p1          UP    ...    # physical port 1
enp3s0f0s0  UP    ...    # a "sub-function" (SF) on the p0 side
enp3s0f1s0  UP    ...    # a "sub-function" (SF) on the p1 side
```

INFO — what is a sub-function? A **sub-function (SF)** is a lightweight virtual NIC carved out of a physical port. It shows up as its own network device. We create one SF on each side and use the two of them as the *endpoints* of a network flow — so a single card can play both “sender” and “receiver” across the cable.

The traffic we will watch is **RoCE** (RDMA over Converged Ethernet), the high-speed, kernel-bypass transport used in AI and storage networks. RoCE does not use normal sockets; programs reach it through **RDMA devices** named `mlx5_0`, `mlx5_1`, `mlx5_2`, `mlx5_3`. List them:

```
$ rdma link show
link mlx5_0/1 ... netdev pf0hpf    # RDMA device for physical port p0
link mlx5_1/1 ... netdev p1        # RDMA device for physical port p1
link mlx5_2/1 ... netdev enp3s0f0s0 # RDMA device for the SF on p0 ← we use this one
link mlx5_3/1 ... netdev enp3s0f1s0 # RDMA device for the SF on p1 ← and this one
```

Remember this mapping: **mlx5_2 and mlx5_3 are the two SF endpoints** we send RoCE between.

mlx5_0/mlx5_1 are the physical ports themselves.

Step 1: wire up the two endpoints. The two ports are wired into a loopback and split into two isolated network sandboxes (Linux network namespaces) called ns0 and ns1 — one SF in each, each with its own IP address (mlx5_2 → ns0 → 10.0.0.1, mlx5_3 → ns1 → 10.0.0.2). This is already set up for you on the tutorial card, so there’s nothing to run here.

INFO — what is a network namespace? It is a private, isolated network stack inside one Linux machine — its own interfaces, IPs, and routes. Putting each SF in its own namespace (ns0, ns1) is what makes them behave like two separate hosts even though they live on the same card. You run a command “inside” a namespace with `ip netns exec <name> <command>`.

Step 2: put real traffic on the loopback. We generate traffic with `ib_write_bw` — a standard RoCE benchmarking tool (from the `perftest` package) that measures how fast one endpoint can write data to another. It needs a server (receiver) and a client (sender).

Try it yourself! Send RoCE across the cable and measure it!

Open **two terminals**, both on the Arm cores.

Terminal 1 — the receiver (server). Start this one first; it waits for a client:

```
sudo ip netns exec ns0 ib_write_bw -d mlx5_2 -R -x 1 -F --report_gbits
```

What the flags mean: `ip netns exec ns0` runs it inside the ns0 sandbox; `-d mlx5_2` uses that namespace’s RDMA device; `-R` sets the connection up via the RDMA connection manager (keep this on — Part II needs it); `-x 1` picks the RoCEv2 address; `-F` ignores a CPU-frequency warning; `--report_gbits` prints the result in gigabits/second. It prints its settings and then says it is **waiting for a client**.

Terminal 2 — the sender (client). Point it at the server’s IP, 10.0.0.1:

```
sudo ip netns exec ns1 ib_write_bw -d mlx5_3 -R -x 1 -F 10.0.0.1 --report_gbits
```

You should see a results table appear on both terminals, with the throughput climbing toward the card’s line rate:

#bytes	#iterations	BW peak[Gb/sec]	BW average[Gb/sec]	MsgRate[Mpps]
65536	529037	0.00	92.46	0.176344

That ~92 Gb/s is your proof the whole path works end to end: sender → p1 → cable → p0 → the eSwitch → receiver. If you see a table with a real number, Part A is done.

To avoid retyping the flags, the repo wraps these as scripts: `./scripts/run_server.sh` and `./scripts/run_client.sh` (one per terminal), or `./scripts/benchmark.sh`, which starts both ends together in a single command and streams the sender’s throughput (Ctrl-C stops both). We use `benchmark.sh` from Part C on.

Part B — DOCA Flow in one page

DOCA Flow is how you program that switch from C. Your program builds a small graph of **pipes**, and each pipe answers four questions:

Part	The question it answers	Example
match	<i>which</i> packets does this pipe act on?	“all IPv4 packets”
monitor	<i>how many</i> packets hit it?	a hardware counter you can read
actions	<i>what</i> do we change in the packet?	rewrite a header field, or nothing
forward	<i>where</i> does the packet go next?	another pipe, a port, the CPU, or drop

You describe these rules once, at startup. The NIC then applies them to every packet in hardware, while your program sits idle printing counters. That is the whole idea: **match, count, modify, forward**.

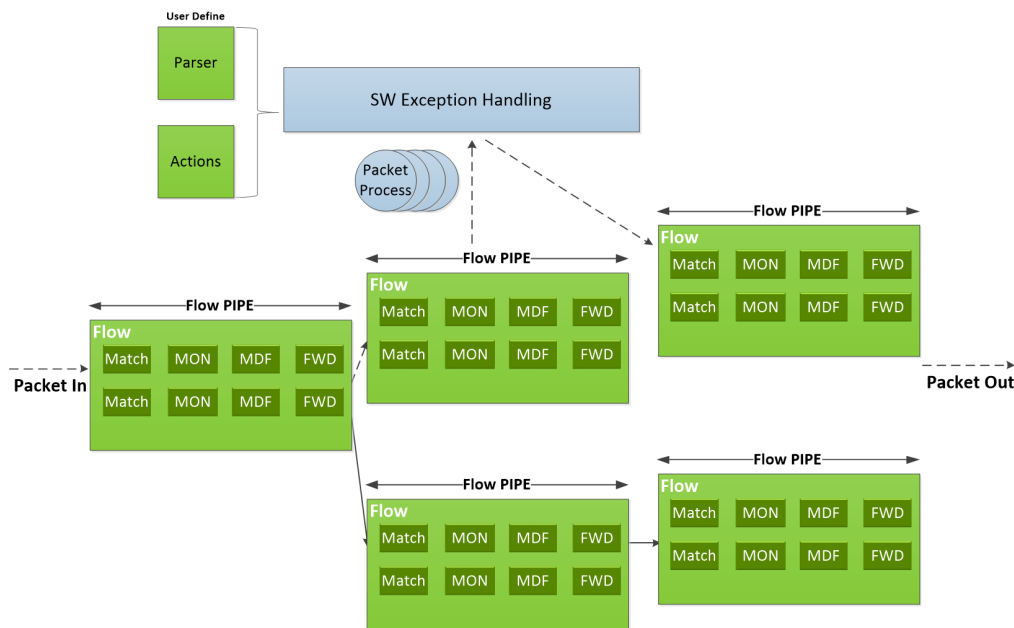


Figure 2: Pipes chain into a graph. Within a pipe, each entry is a Match, a monitor (MON), a modify/action stage (MDF) and a forward (FWD). Source: [NVIDIA DOCA Flow programming guide](#), “Architecture”.

Two things about pipes are worth getting straight before you write any, because both are easy to get backwards:

Pipe = which fields. Entry = what values. A pipe is a *template*: writing `0xFF` into a field of its match means “this field participates”, and `0x00` in the mask means “...but do not actually compare it”. The **entry** you add afterwards supplies the value really compared. Creating a pipe installs no rule in the hardware; adding an entry does. The same split applies to actions — the pipe declares “entries may rewrite this field”, the entry says “...to this value”.

One pipe is the root. Every packet entering the switch starts its lookup at the pipe marked `is_root`, and that is the only way into the rest of the graph. Until the root pipe exists, none of your other pipes are reachable no matter how correct they are — which is exactly why the untouched program forwards nothing.

You start from two worked examples. In `doca_flow_ecn_pcap.c`, `create_passthrough_pipe()` is a complete, minimal pipe and `create_to_cpu_pipe()` shows how to deliver packets to your own process. Neither needs changing — read the first one before you write anything, because **every** pipe in the file, including all three of yours, has the same five-part shape:

1. **Create a pipe configuration** with `doca_flow_pipe_cfg_create()`, against the port it belongs to. What comes back is a builder, not a pipe.
2. **Describe it**, with one setter per property: `..._set_name()`, `..._set_type()`, `..._set_domain()`, `..._set_is_root()` and `..._set_nr_entries()`, plus `..._set_match()` for the match template — and `..._set_actions()` and `..._set_monitor()` when the pipe rewrites or counts. `..._set_match()` takes two `doca_flow_match` structs: the first says which fields participate, the second masks them.
3. **Create the pipe** with `doca_flow_pipe_create()`, passing two `doca_flow_fwd` structs — where a hit goes, and where a miss goes. Then discard the builder with `doca_flow_pipe_cfg_destroy()`; the pipe keeps no reference to it.
4. **Add the entries** with `doca_flow_pipe_add_entry()`, once each. This is the call that actually installs a rule — the pipe on its own does nothing.

5. **Install and check** with `doca_flow_entries_process()`, then confirm that every entry landed.

Step 5 is not optional. `doca_flow_pipe_add_entry()` returns as soon as the driver has taken the rule; the hardware's verdict arrives later, through a callback. A pipe that exists but installed nothing forwards nothing, silently — the hardest failure here to spot from the outside.

Part C — Build the pipeline

In `doca-2/doca-flow/doca_flow_ecn_pcap.c`, three function bodies are empty, marked `TODO 1` to `TODO 3`. Everything else is done, and the program as shipped already forwards traffic. Laid out logically, the whole program is three phases, and only the middle one is yours (we use Python here purely as pseudo-code, to show the logical components of the template C file we provide you):

```
def main():
    setup_logging()
    parse_args()                # --pcap, --percent, --sample
    open_capture_pcap()         # only if --pcap was given

    # ---- bring-up: device and library. None of this is pipeline work ----
    dev = open_and_probe_dev(0) # PF0, probed into DPDK with its SF representor
    configure_and_start_dpdk_port(dev) # packet buffer pool, RX/TX queues
    initialize_doca_flow()        # switch mode, hardware steering, counters
    port = port_start(dev)        # the PF0 proxy port -- must be started first
    sf = rep_port_start(...)      # the server SF's representor

    build_pipeline(port, cfg, out) # <-- ALL of your work is reached from here

    # ---- runtime ----
    run_capture_loop()           # populates the pcap & print counters
    teardown()
```

Those are the real function names, so you can `grep` for any of them.

`build_pipeline()` is the one to read closely: it is the whole exercise in a single function. Note that it ships wired as a **plain forwarder** — the commented-out half is what you are here to build.

```
static void build_pipeline(struct doca_flow_port *port, const struct app_config *cfg,
                          struct pipeline *out) {
    // Both configurations need this pipe. In the no-op it IS the data path; in the ECN pipeline
    ↪ it is
    // where the two marking pipes send whatever they do not match.
    struct doca_flow_pipe *passthrough = create_passthrough_pipe(port);

    // ----- NO-OP CONFIGURATION: comment out this line for the exercise.
    ↪ -----
    create_root_pipe(port, passthrough);

    // ----- ECN CONFIGURATION: uncomment everything below.
    ↪ -----
    //
    // bool capture = (cfg->pcap_path != NULL);
    //
    // if (capture) {
    //     struct doca_flow_pipe *cpu = create_to_cpu_pipe(port);
    //     bind_capture_mirror(port, cpu);
    // }
    //
```

```

// // PASS_CAPTURE (no mark) and MARK_CAPTURE (CE-mark); both mirror to pcap only when
↪ capturing.
// struct doca_flow_pipe *pass_cap =
//     create_forward_to_sf_pipe(port, false, capture, passthrough, &out->pass_entry);
// struct doca_flow_pipe *mark_cap = NULL;
// if (cfg->random_percent > 0.0)
//     mark_cap = create_forward_to_sf_pipe(port, true, capture, passthrough, &out->ce_entry);
//
// // wire-ingress entry point per --percent
// struct doca_flow_pipe *wire_target;
// if (cfg->random_percent >= 100.0)
//     // mark+capture all
//     wire_target = mark_cap;
// else if (cfg->random_percent <= 0.0)
//     // capture all, mark none
//     wire_target = pass_cap;
// else {
//     out->sample_mask = get_random_mask(cfg->random_percent);
//     wire_target = create_sampling_pipe(port, mark_cap, pass_cap, out->sample_mask);
// }
//
// create_root_pipe(port, wire_target);
}

```

[3.x] Two differences. `create_flood_pipe()` replaces `bind_capture_mirror()` as TODO 2, and is built *after* passthrough because it forwards to it. And TODO 1 takes that flood pipe where 2.x takes a capture flag.

Part D — Tutorial exercise

Three parts. The first two take a few minutes each; the third is the actual programming.

D.1 — Run it, and watch it forward

Build and run the template exactly as it comes, with no changes at all:

```

$ meson setup doca-2/build doca-2
$ ninja -C doca-2/build
$ sudo ./doca-2/build/doca-flow/doca_flow_ecn_pcap -- --percent 100

```

Your first build prints about five defined but not used warnings. That is expected: the ECN half of `build_pipeline()` is commented out, so the functions it would call are not referenced yet. They clear as you uncomment in D.3.

With `./scripts/benchmark.sh` running in a second shell, **traffic should flow at line rate** — 92 or 184 Gb/s depending on the card — and the counter line should sit at zero:

CE marked: 0, passthrough: 0 (0% marked)

That is the no-op configuration doing exactly what it says: every packet from the wire is handed to the server untouched, everything coming back goes out to the wire, nothing is marked or counted.

Now stop it with Ctrl-C, leaving the traffic running. Throughput carries on unchanged — the card falls back to its default OVS forwarding. Start the program again and it takes over. That switch, default path versus *your* path, is the control you are about to use.

Why this matters. The instant a DOCA Flow program starts it **owns the NIC's switch**, and from then on the NIC forwards only what your pipes say to forward. The template is careful to install a working forwarder before it does anything else. Delete that and noth-

ing moves at all: an empty pipeline does not mean “pass traffic through”, it means “drop everything”.

D.2 — Switch the configuration over

Open `doca-2/doca-flow/doca_flow_ecn_pcap.c` and find `build_pipeline()`. Make two edits:

1. **Comment out** the single line of no-op wiring:

```
// create_root_pipe(port, passthrough);
```

2. **Uncomment** the ECN configuration block underneath it.

Rebuild and run it. **It will not start**, and the last line of the error will name `PORT_DEMUX`:

```
$ ninja -C doca-2/build
$ sudo ./doca-2/build/doca-flow/doca_flow_ecn_pcap -- --percent 100
```

That is correct, and it is why this step is worth doing on its own. The ECN configuration calls `create_forward_to_sf_pipe()`, which is still a `TODO` returning `NULL`, and the root pipe is then asked to forward wire traffic into a pipe that does not exist. You have taken the working forwarder away and not yet put anything in its place; traffic stops either way. D.3 is where you put it back.

D.3 — Build the ECN pipeline

Three `TODO`s, in this order. Only the first is needed to see something work.

`TODO 1` — `create_forward_to_sf_pipe()`, *the marking*

The heart of the exercise. It is called **twice** — once with `mark` false, once true — so everything mark-specific goes behind `if (mark)`. Start from `create_passthrough_pipe()` and add:

- **Match** any IPv4 packet *whatever ECN bits it arrived with*, using the wildcard idiom from Part B: `outer.ip4.dscp_ecn` as `0xFF` in the pipe’s match, `0x00` in the mask. Reset that byte to `0x00` before adding the entry — the same struct is reused as the entry’s values.
- **A counter**, via `monitor.counter_type = DOCA_FLOW_RESOURCE_TYPE_NON_SHARED`. Without it the CE marked: line stays at zero and you cannot tell whether anything works.
- **The marking action**, when `mark` is true: declare `outer.ip4.dscp_ecn` as `0xFF` in a pipe-level action template, and have the entry write the value. RFC 3168 gives the two-bit ECN field as Not-ECT 00, ECT(1) 01, ECT(0) 10, CE 11, so the byte you want is `0x03`.
- **Forwards**: hits to the server’s SF, misses to `miss_pipe`.
- Hand the installed entry back through `out_entry` — that is what the counter report queries.

Rebuild and run with `--percent 100` and no `--pcap`, so the other two `TODO`s are never reached:

```
$ sudo ./doca-2/build/doca-flow/doca_flow_ecn_pcap -- --percent 100
```

Traffic should be back at line rate, and now the counter climbs:

CE marked: 57060637, passthrough: 0 (100% marked)

That is you rewriting headers in hardware. You cannot *see* the bit yet — the counter only proves packets went through your marking pipe. That is what `TODO 2` is for.

`TODO 2` — *the capture copy*

Sends a *copy* of every packet to a file while the original carries on to the server untouched.

[2.x] `bind_capture_mirror()` builds no pipe. A mirror is a port-level *shared resource*: point its target’s

fwd at cpu_pipe — **this is where the copy goes** — then use DOCA_TUT_MIRROR_SET_ORIG_FWD for where the **original** carries on, then configure it under MIRROR_ID and *bind* it to the port. Configuring alone does nothing.

[3.x] create_flood_pipe() builds a DOCA_FLOW_PIPE_HASH pipe running the **flooding** algorithm, which delivers each packet to *all* of its entries instead of hashing it to one. Exactly two entries, since a hash pipe’s count must be a power of two: entry 0 to production_pipe, entry 1 to capture_pipe. Ordering is only guaranteed for entry 0, so the real data path goes there and the copy takes entry 1. Clear the forward struct between the two. The constant is DOCA_FLOW_PIPE_HASH_MAP_ALGORITHM_FLOODING.

Now run with a capture file. Writing starts **paused**:

```
$ sudo ./doca-2/build/doca-flow/doca_flow_ecn_pcap -- --pcap /tmp/out.pcap --percent 100
```

CE marked: 57060637, passthrough: 0 (100% marked) | mirrored: 2743653 -> pcap: 0 [PAUSED]

mirrored: (**[3.x]**: flooded:) climbing means copies are reaching the CPU. Press **SPACE** to start writing, wait a few seconds, then Ctrl-C to flush and close the file. Then look at your mark:

```
$ ./scripts/check_ecn_bits_from_pcap.sh /tmp/out.pcap
#   tos 0x3,CE   <- a marked packet
#   (no tos)     <- an unmarked one; tcpdump omits the field when the byte is zero
```

At --percent 100 every IPv4 packet should read CE, and throughput should have stayed at line rate throughout — the copy is made in hardware.

TODO 3 (optional) — *mark only some packets*

create_sampling_pipe() splits traffic probabilistically, in hardware. The NIC stamps every packet with a random 16-bit value in parser_meta.random; match it against 0 under mask — already computed for you as a power of two minus one — and exactly 1 packet in (mask + 1) hits. Hits go to the marking pipe, misses to the non-marking one; “miss” here means *not selected*, not an error.

```
$ sudo ./doca-2/build/doca-flow/doca_flow_ecn_pcap -- --pcap /tmp/out.pcap --percent 50
```

The startup banner prints the fraction actually achieved, rounded down to a power of two. Check it against the mix of marked and unmarked packets in the capture.

Check your answer

The finished program sits next to yours:

```
$ diff doca-2/doca-flow/doca_flow_ecn_pcap.c doca-2-solutions/doca-flow/doca_flow_ecn_pcap.c
```

Expect your three function bodies, plus build_pipeline() — the solution has the ECN configuration live and no no-op wiring, which is exactly the edit you made in D.2.

Debugging tips

- **Read the *last* error line, not the first.** A failed pipe prints a wall of internal DOCA errors. The final [CRT]...[doca_check] <name>: <reason> line names the pipe, and that name is the function to open.
- **It runs, but nothing forwards** — expected between D.2 and D.3: the root pipe is aimed at create_forward_to_sf_pipe(), which returns NULL until you write it. If it persists after TODO 1, check that pipe’s forwards.
- **CE marked: stays at 0** — your marking pipe has no counter (monitor.counter_type), or you are

still running the no-op configuration; check you commented out that line in D.2.

- **Throughput collapses** — the data path is going somewhere it should not. Re-check the forwards in `create_forward_to_sf_pipe()`.
- **The pcap stays empty** — capture starts **paused**; press SPACE. That needs a real terminal, so run the program in the foreground.
- **A new run says the device is busy** — only one DOCA Flow program can own the switch at a time. Stop the previous one; if it was killed hard, `sudo pkill -f doca_flow`.
- **A pipe fails to install** — check the status after `doca_flow_entries_process()`, per step 5 of the shape in Part B. Success from the add-entry call alone proves nothing.

What you built

You can now treat the card as a two-port loopback carrying real RoCE traffic; follow how match/count/-modify/forward pipes chain from a root pipe; and write a pipeline that matches IPv4, counts it, rewrites its ECN bits to CE, forwards it, and copies it to a capture file — all in hardware, at line rate, with your program idle. That CE mark is exactly the signal the congestion controller in Part 2 reacts to.

Appendix A — The BlueField, in more detail

Reference material. You do not need any of this to finish the exercise.

Interacting with a BlueField-3 typically entails interfacing with — and often independently programming — four different architectural components:

Where	What it is	Used for
Host x86	The server the card is plugged into	Not used at all in this tutorial
Arm cores	Cortex-A78 cores running their own Ubuntu on the card	Where you log in, build and run
NIC ASIC / eSwitch	The match-action switch inside the NIC	What you program in Part 1
DPA	Datapath Accelerator: a many-threaded RISC-V engine on the data path	The congestion controller, Part 2

The cards are configured in **DPU mode** (also called embedded-CPU-function ownership, ECPF): the Arm subsystem owns the NIC's resources, and the host x86 sees only a plain network device.

Functions: PFs, VFs and SFs

A **function** is what the NIC exposes so that software can send and receive packets through it. It is not a piece of software, but a slice of the NIC hardware, with its own queues, its own MAC address, and its own identity on the network. There are three kinds of functions:

- **PF** (*physical function*): the real PCIe device the NIC presents. A BlueField-3 has one PF per physical port. These are the functions that own the hardware.
- **VF** (*virtual function*): an [SR-IOV](#) slice of a PF, created so that a virtual machine can be handed something that looks like its own NIC. Not used in this tutorial.
- **SF** (*scalable function*, also called a *sub-function*): the same idea as a VF, but not tied to PCIe. An SF is created on the Arm with `m1nx-sf`, is much cheaper than a VF, and you can have far more of them. **This tutorial uses SFs** as its traffic endpoints.

Each function shows up on the Arm's Linux as **two** different devices:

Interface	What it is	Example
netdev	An ordinary Linux interface — what <code>ip link</code> lists and <code>ping</code> uses.	<code>p0</code>
RDMA device	The same function through the RDMA stack, bypassing the kernel.	<code>m1x5_0</code>

In this tutorial we use RDMA over Converged Ethernet (**RoCE**), which is RDMA carried inside ordinary UDP/IP packets, so we use the RDMA devices (`m1x5_N`). Because RoCE is just UDP/IP on the wire, the switch inside the NIC can match and rewrite it like any other packet.

Function	RDMA device	Netdev
PF0	<code>m1x5_0</code>	<code>p0</code>
PF1	<code>m1x5_1</code>	<code>p1</code>
SF on PF0	<code>m1x5_2</code>	<code>enp3s0f0s0</code>
SF on PF1	<code>m1x5_3</code>	<code>enp3s0f1s0</code>

The eSwitch

The eSwitch (embedded switch) is a hardware block inside the NIC that forwards packets between every function and every physical port on the card. Every packet that arrives from the wire and every packet

a function transmits passes through it, and it decides where that packet goes. It is the single most important component in this part of the tutorial, and we use DOCA Flow to program it.

Its rule table is called the **FDB** (forwarding database). Each PF (port) owns its own logical eSwitch domain, with its own rules, so programming PF0’s eSwitch leaves PF1’s forwarding untouched.

Every endpoint the eSwitch can send a packet to is a **vport** (virtual port): the physical ports are vports, and so is every PF, VF and SF. The eSwitch does not reach an SF directly, though. It reaches the SF’s **representor** — a switch-side netdev that stands in for it. The two are the same function seen from opposite sides:

- from *inside* the namespace or VM using it, the SF appears as its netdev and its `mlx5_N` RDMA device — this is where an application sends and receives;
- from the *switch*’s side, the same SF appears as its representor, which is what a forwarding rule names.

So “deliver this packet to the receiver” is written, in a rule, as “output to that representor’s vport” — and the packet then pops out of the corresponding `mlx5_N` device inside the namespace. In NVIDIA’s figure below, `rep0` and `rep1` are the representors of `VF0` and `VF1`; substitute SFs for VFs and it is our layout exactly, with the DOCA Flow application running on the Arm.

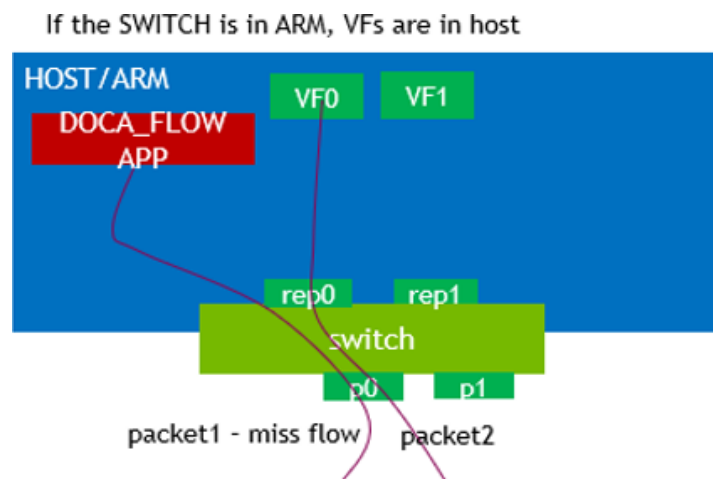


Figure 3: Applications use the functions (`VF0`, `VF1`); forwarding rules name their representors (`rep0`, `rep1`) and the physical ports. Source: [NVIDIA DOCA Flow programming guide](#), “Domains in Switch Mode”.

By default, each PF’s vports are wired together by an OVS bridge with hardware offload — the “normal” forwarding path that makes the card behave like an ordinary NIC when nothing else is running. A DOCA Flow program **replaces** that for the PF it attaches to, from the moment it starts until it exits.

The whole round trip

The client issues an RDMA WRITE through `mlx5_3`; PF1’s eSwitch — untouched by you — sends it out `p1`; it crosses the cable to `p0`; **PF0’s eSwitch, your pipeline, marks and forwards it**; the server receives it via `mlx5_2`; the server’s NIC answers a CE-marked packet with a **CNP** (Congestion Notification Packet) in hardware; the CNP travels back the same way; and on the client’s side it raises an event on the DPA, where the Part 2 algorithm sets a new send rate.

Appendix B — DOCA Flow concepts in full

Reference material. Part B has the working subset.

Port. A DOCA Flow handle on one vport. In *switch mode* the PF uplink is also the **proxy port** (switch-manager port): it must be started first, and every pipe belongs to it — even pipes that forward somewhere else entirely.

DOCA Flow + DOCA PCC (USER_PROGRAMMABLE_CC=1, PCC loaded)

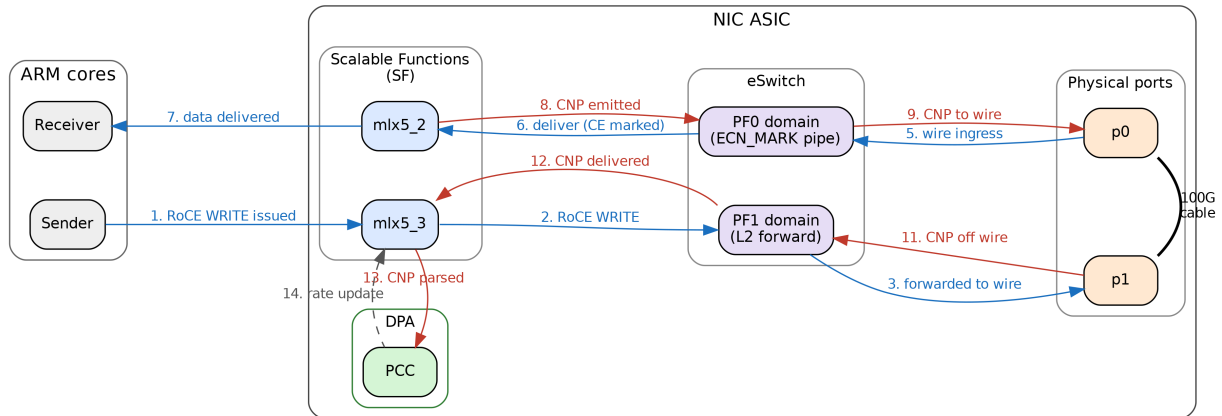


Figure 4: The end-to-end data path with both parts running.

Pipe. A flow table, not a pipeline stage. NVIDIA’s wording: a pipe “is a template that defines packet processing without adding any specific hardware rule”.

Pipe type	What it does
BASIC	Ordinary match-action table
CONTROL	Entries carry priorities (0-7), resolving conflicts between overlapping matches
LPM	Longest-prefix match, for routing-table-shaped lookups
ACL	Five-tuple matching with masks
ORDERED_LIST	A fixed sequence of actions per entry, when their order matters
HASH	Selects an entry by an index computed from a hash, rather than by matching fields

Entry. A concrete rule inside a pipe.

Match. Each field is *ignored* (left zero), *constant* (set in the pipe, the same for all entries), or *changeable* (all-ones placeholder in the pipe, real value supplied per entry).

Actions. Mostly field rewrites — MAC, IP, DSCP/ECN, L4 ports, metadata — but also encapsulation and decapsulation. Matches and actions are not alternatives to one another: every entry has a match, and *may additionally* have actions, a monitor and a forward.

Monitor. Counters, meters, and (on 2.x) mirroring.

Forward. Where the packet goes when the lookup finishes. Each pipe has a *hit* forward and optionally a *miss* forward:

Forward	Meaning
DOCA_FLOW_FWD_PORT	Output to a vport: a physical port, or a function’s representor
DOCA_FLOW_FWD_PIPE	Jump to another pipe — this is how pipes chain
DOCA_FLOW_FWD_RSS	Deliver to a receive queue of your program, on the Arm cores
DOCA_FLOW_FWD_DROP	Discard
DOCA_FLOW_FWD_CHANGEABLE	Each entry brings its own forward
DOCA_FLOW_FWD_HASH_PIPE	Jump to a hash pipe, naming the algorithm to run (3.x)

Shared resources. Objects living on the port rather than inside one pipe, so several pipes can point at a single instance: meters, counters, RSS contexts, encap/decap contexts — and, on 2.x, **mirrors**. Mirrors were removed in DOCA 3.2, which is the largest difference between the two versions of this exercise.

Parser metadata. `parser_meta` holds values the hardware parser attaches to each packet, rather than header fields: `outer_l3_type` (what the parser saw), `random` (a fresh 16-bit value per packet, which is what makes hardware sampling possible), and the ingress port.

Note that `outer.ip4.dscp_ecn` is the **whole** TOS byte, so writing `0x03` sets ECN to CE *and* clears DSCP. Our traffic carries DSCP 0, so it makes no difference here.

Appendix C — The API calls used in this exercise

The headers on the card are the authority: `/opt/mellanox/doca/include/doca_flow.h`, with packet field types in `doca_flow_net.h`. Every call is documented there. This is only a map of which ones matter here.

`grep -n 'dscp_ecn\|parser_meta' /opt/mellanox/doca/include/doca_flow*.h` answers most structural questions faster than the web documentation. If you are on VS Code Remote-SSH, `.vscode/c_cpp_properties.json` already points IntelliSense at these paths, so “go to definition” works.

Call	What it is for
<code>doca_flow_pipe_cfg_create / _destroy</code>	Start and finish describing a pipe
<code>doca_flow_pipe_cfg_set_name</code>	Name it — this is what error messages report
<code>doca_flow_pipe_cfg_set_type</code>	BASIC, HASH, ...
<code>doca_flow_pipe_cfg_set_domain</code>	Which steering domain; always <code>..._DOMAIN_DEFAULT</code> here
<code>doca_flow_pipe_cfg_set_is_root</code>	Mark the one root pipe
<code>doca_flow_pipe_cfg_set_nr_entries</code>	How many entries you will add
<code>doca_flow_pipe_cfg_set_match</code>	The match template and its mask
<code>doca_flow_pipe_cfg_set_actions</code>	The action template(s)
<code>doca_flow_pipe_cfg_set_monitor</code>	Counter, meter, and [2.x] the mirror id
<code>..._cfg_set_hash_map_algorithm</code>	[3.x] flooding, for the capture copy
<code>doca_flow_pipe_create</code>	Create the pipe, with its hit and miss forwards
<code>doca_flow_pipe_add_entry</code>	Add an entry — [2.x] and 3.1
<code>doca_flow_pipe_basic_add_entry</code>	Add an entry — [3.x] ; takes the action index as an argument
<code>doca_flow_pipe_hash_add_entry</code>	Add an entry to a hash pipe, by index
<code>doca_flow_entries_process</code>	Drive queued entries to completion, then check the status
<code>doca_flow_resource_query_entry</code>	Read an entry’s counter (already written, in <code>query_pkts()</code>)
<code>doca_flow_shared_resource_set_cfg</code>	[2.x] configure the mirror
<code>doca_flow_shared_resources_bind</code>	[2.x] bind it to the port — configuring alone does nothing

Version differences

	[2.x]	[3.x]
copy to the	shared mirror, via	<code>DOCA_FLOW_PIPE_HASH +</code>
pcap	<code>monitor.shared_mirror_id</code>	<code>..._ALGORITHM_FLOODING</code>
entry	<code>doca_flow_pipe_add_entry</code> ; action index	<code>doca_flow_pipe_basic_add_entry</code> ;
install	inside <code>doca_flow_actions</code>	action index as an argument
entry flags	<code>DOCA_FLOW_NO_WAIT</code> , <code>DOCA_FLOW_WAIT_FOR_BATCH</code>	<code>DOCA_FLOW_ENTRY_FLAGS_NO_WAIT</code> , <code>..._WAIT_FOR_BATCH</code>
RSS	flat <code>fwd.rss_queues / num_of_queues /</code>	nested <code>fwd.rss_queues_array /</code>
forward	<code>rss_outer_flags</code>	<code>nr_queues / outer_flags</code>
ingress port	<code>parser_meta.port_meta</code> (uint32)	<code>parser_meta.port_id</code> (uint16)
match		

`doca_flow_compat.h`, force-included by the build, already smooths the 3.1-versus-3.4 seam inside the `doca-3` tree. It does **not** bridge 2.x to 3.x.

The program's own flags

Flag	Meaning
--percent N	CE-mark this share of packets, $[0, 100]$, rounded down to a power-of-two fraction. Default 100.
--pcap FILE	Also capture a copy of the traffic to FILE. Omit for pure marking mode.
--sample N	Write only about 1-in-N captured packets to the file. Marking and forwarding are unaffected.

Further reading

- [DOCA Flow programming guide](#) — swap the version in the URL to match your card, e.g. .../archive/2-9-0/doca-flow.
- [BlueField modes of operation](#)
- [BlueField scalable function user guide](#)
- NVIDIA ships around 20 single-concept sample programs on the card. They live under /opt/mellanox/doca/samples/doca_flow/ — one .c file plus a README each, every one isolating a single idea.